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Loading arrangement for loading and unloading load units on and off a vehicle

Abstract

A loading arrangement for loading and unloading load units on and/or off a vehicle comprises a frame, wherein said frame has first and second ends and to be fastened to the vehicle in a detachable manner. The arrangement comprises also a transferring device for catching and transferring the load unit on and/or off the vehicle, said transferring device comprising a gripping point to be gripped by a hook system of the vehicle and a catching device for catching said load unit and for transferring it on or off the vehicle, when the transferring device is operated by said hook system of the vehicle. The frame comprises a dock, which is arranged in the first end of the frame and wherein the dock comprises a receiving portion for receiving the transferring device for at least loading and unloading said frame on and/or off the vehicle by the transferring device when said transferring device is operated by a hook system of the vehicle.

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Background/Summary

CROSS REFERENCES

(1) This application is a U.S. national stage application of international patent application number PCT/FI2021/050466 filed on Jun. 18, 2021 claiming priority to Finnish national application number FI20205644 filed on Jun. 18, 2020.

TECHNICAL FIELD OF THE INVENTION

(2) The invention relates to a loading arrangement for loading and unloading load units on and/or off a vehicle, such as a truck.

BACKGROUND OF THE INVENTION

(3) It is known to transport a skip or interchangeable load unit by a vehicle, such as a truck, and load and unload the load unit on or off the vehicle using a skip arrangement. In order to enable the loading and unloading of the load unit by the skip arrangement, the vehicle need to be provided typically for example with a hook system, but in addition to the load unit must be compatible with the skip arrangement of the vehicle and the vehicle must be configured to receive the load unit with the compatible skip arrangement.

(4) However, some problems may arise due to different types of load units, which are not compatible with the skip arrangements of the vehicle, whereupon this kind of load units, such as for example sea containers or shipping containers, cannot be pulled by the hook system of the vehicle on the vehicle. Also, some problems may arise due to different hook systems or skip arrangements of the vehicle, which are not compatible with the load units, for example it is not possible to catch and load the sea or shipping container with a conventional hook system or lift system of the vehicle and pull or drag it on the vehicle.

SUMMARY OF THE INVENTION

(5) An object of the invention is to alleviate and eliminate the problems relating to the known prior art. Especially the object of the invention is to provide an arrangement and method for loading and unloading load units on and/or off a vehicle regardless of the type of load unit or vehicle. In addition, the object of the invention is to make possible to load and unload load units on and/or off the vehicle regardless of the type of load unit or vehicle and so that there is no need to make any modifications either to vehicles or load units.

(6) The object of the invention can be achieved by the features of independent claims.

(7) The invention relates to an arrangement for loading and unloading load units on and/or off a vehicle according to claim 1. In addition, the invention relates to a transferring device, a frame, and a method for loading and unloading load units on and/or off a vehicle.

(8) According to an embodiment of the invention a loading arrangement for loading and unloading

load units on and/or off a vehicle comprises a frame and a transferring device. The frame has first and second ends and is configured to be fastened to the vehicle in a detachable manner. The load unit may be for example a container or sea container, shipping container, tank, or the like and has first and second ends. The load unit comprises advantageously also at least a first fastening device so that the load unit can be caught. The first fastening device is for example hole or hook or the like e.g. by sides of the load unit in said first end but can naturally be also other kind of fastening device, such as a pin, pivot, plug, groove or cable. For example, the sea container typically comprises the fastening device in each corner of it, the fastening device having a hole in each side, and in particularly in the bottom portion and side portion (also in the front and rear portions), and both in said first end second ends of the load unit.

(9) The frame comprises advantageously two elongated frame elements with first and second ends, having a longitudinal axis between said ends. In addition, the frame has advantageously an opening between the elongated frame elements so that for example the hook system of the vehicle can be operated between said frame elements in the direction of the longitudinal axis, so backward and forward directions. For example, when the frame is arranged or mounted on the vehicle, the hook system can be lifted up and transferred via or in the opening for loading or unloading the load unit on or off the vehicle so that there is no need to remove the frame from the vehicle.

(10) In addition, according to advantageous embodiment the frame comprises a dock, which is arranged in the first end of the frame. Advantageously the dock is arranged in fixed way, whereupon it also strengthens the structure durability of the frame. In addition, when the dock is arranged in the first end, the opening is still remained between the elongated frame elements so that the hook system of the vehicle can be operated. The dock comprises advantageously a receiving portion for receiving the transferring device. When the transferring device is received by the receiving portion of the dock, the frame can be loaded and unloaded on and/or off the vehicle by operating the transferring device with the hook system of the vehicle. In addition, the receiving portion of the dock may also be used for receiving the first end, such as the bottom end, of the load unit when the load unit is loaded on the vehicle.

(11) The dock, such as the receiving portion, comprises advantageously a locking device to be fastened to the first fastening device, such as the first fastening hole in the underside or bottom portion of the load unit in said first end (so the fastening hole pointing downwards in the sea container, for example). In this way the first end of said load unit can be fastened to the frame in a detachable manner, and again via the frame to the vehicle.

(12) According to an advantageous embodiment the frame comprises also a ramp, which is hinged in the second end of the frame. The ramp is also configured to be hinged around a rear roller of the vehicle. This can be implemented e.g. by at least one separate locking hinge piece, which is arranged around (at least partly around) the rear roller and then fastened to the ramp, advantageously to the second end portion of the ramp. This offer advantage, namely in this way the ramp divides load or stress of the load unit to the vehicle structure via the rear roller of the vehicle, especially when said load unit is transferred (loaded or unloaded). In addition, the frame can be fastened in this way to the vehicle via said rear roller in a detachable manner.

(13) According to an embodiment the transferring device is used for loading and unloading the frame, but also for catching and transferring the load unit on and/or off the vehicle. The transferring device comprises advantageously a catching device for catching the load unit via the fastening device of the load unit. The catching device may be for example a hook, pin, pivot, plug, clamp, bolt, twist, screw or cable or the like, depending on the type of the fastening device of the load unit. For example, the fastening device of the typical sea container is a hole (at least two holes pointing different sides in each corner, so first and second fastening devices), whereupon the catching device is advantageously a plug or other suitable catching device, which can be used for catching.

According to an embodiment, the catching device of the transferring device is replaceable catching device.

(14) In addition, the transferring device advantageously comprises a gripping point to be gripped by the hook system of the vehicle. Thus, when the catching device is fastened to the fastening device of the load unit, the load unit can be transferred, such as loaded or unloaded on or off the vehicle by the transferring device, when the transferring device is operated by the hook system of the vehicle. It is to be noted that in this way also the typical sea container can be loaded to the vehicle with the conventional type skip system or hook lift system.

(15) The transferring device may also comprise an elongated boom having first and second ends, where the elongated boom is coupled with the transferring device via a joint arranged in the second end of the elongated boom. In this embodiment, the elongated boom comprises also the gripping point to be gripped by the hook system of the vehicle. The gripping point is advantageously arranged next to the joint to the second end of the elongated boom and towards the first end of the elongated boom from the gripping point. Thus, when the joint locates closer to the second end of the elongated boom than the gripping point, it functions as a levelling point and balances (due to gravity) a position and possible swinging of the transferring device when it is moved e.g. towards the load unit for catching it. In addition, the elongated boom functions as a lever arm and supports the first end (bottom portion of it) of the elongated boom to the boom of the hook system of the vehicle during loading/unloading the load unit by the transferring device. This allows more optimal position and angle to the hook system for lifting the load unit during loading/unloading.

(16) According to an embodiment the frame is configured to receive and secure the transferring device in a detachable manner. In addition, the dock may comprise a securing device for securing the transferring device in a detachable manner. The securing device may comprise for example a hook to be fastened to a corresponding groove in the transferring device, but naturally the securing device and the corresponding securing point in the transferring device can be implemented also in other ways, such as by a rim screw or the like. Advantageously the transferring device is configured to be released for transferring the load unit on and/or off the vehicle and to be secured for transporting said transferring device, for example. In addition, it is to be noted that the transferring device is advantageously secured to the dock of the frame for loading and unloading the frame, where the transferring device is operated by the hook system and thereby also the frame to which said transferring device is coupled is transferred.

(17) The embodiments of the invention offer advantages, namely when the catching device of the transferring device is unfastened from the second fastening device of the load unit and the locking device of the dock is still fastened to the first fastening device of the load unit, the transferring device can be removed from the dock and frame so that the load unit is still fastened to the dock and frame via the locking device of the dock. If the frame is also fastened to the vehicle, the load unit is also fastened to the same vehicle via **10** the frame, or the frame with the load can also be left to the ground and the transferring device can be taken to the vehicle for example for loading another frame (with or without) to the vehicle, for example.

(18) In addition, it is to be noted that the frame can be unfastened from the rear roller of the vehicle, whereupon the frame with the load unit can be transferred to a ground by the transferring device (when the transferring device is still secured to the dock and frame). When the frame with the load unit is transferred to the ground, the transferring device can be then unfastened from said second fastening device and removed so that the frame with the load unit is still left to the ground. In addition, it is to be noted that the transferring device can be used for transferring the frame with the fastened load unit also from the trailer to the vehicle and vice versa, from the vehicle to the trailer.

(19) The invention also offers further advantages, namely the embodiments of the invention described in this document can be used for loading and unloading the load units on and/or off the vehicle regardless of the type of the load unit or vehicle in an easy, safe and fast way. In addition, according to the embodiments of the invention there is even no need to make any modifications either to vehicles or load units for enabling the loading and unloading of the load units on and/or

off the vehicle regardless of the type of the load unit or vehicle. In other words, there is no need to weld any rails or bars on the bottom portion of the sea container in order to make sea container compatible with the vehicles with the hook lift skip systems. In addition, if the sea container would be provided with the fixed rails or bars, it would not be seaworthy anymore. Still in addition, any vehicle with a suitable hook lift system (small or bigger) can be turned a vehicle for transporting e.g. sea or shipping containers so that there is again no need to make any modifications to the vehicle or load units, such as sea or shipping containers, as such. Furthermore, because there is no need to provide the vehicles or the load units with fixed modifications, lots of savings can be achieved, because there is no additional mass to be transported.

(20) The exemplary embodiments presented in this text are not to be interpreted to pose limitations to the applicability of the appended claims. The verb “to comprise” is used in this text as an open limitation that does not exclude the existence of also unrecited features. The features recited in depending claims are mutually freely combinable unless otherwise explicitly stated.

(21) The novel features which are considered as characteristic of the invention are set forth in particular in the appended claims. The invention itself, however, both as to its construction and its method of operation, together with additional objects and advantages thereof, will be best understood from the following description of specific example embodiments when read in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Next the invention will be described in greater detail with reference to exemplary embodiments in accordance with the accompanying drawings, in which:

(2) FIG. 1 illustrates an exemplary vehicle with a loading arrangement for loading and unloading load units on and/or off a vehicle according to an advantageous embodiment of the invention,

(3) FIGS. 2-6 illustrate a principle of an exemplary transferring device of the loading arrangement according to an advantageous embodiment of the invention,

(4) FIGS. 7-38 illustrate a principle of an operation of an exemplary loading arrangement for loading and unloading load units on and/or off a vehicle according to an advantageous embodiment of the invention,

(5) FIGS. 39-49 illustrate a principle of an operation of an exemplary frame for loading and unloading load units on and/or off a vehicle according to an advantageous embodiment of the invention,

(6) FIGS. 50-57 illustrate example embodiments of the loading arrangement according to an advantageous embodiment of the invention,

(7) FIG. 58-61 illustrates exemplary transferring devices according to an advantageous embodiment of the invention,

(8) FIGS. 62-66—illustrate an example method and embodiments of the loading arrangement according to an advantageous embodiment of the invention,

(9) FIGS. 67-68—illustrate still an additional view of the embodiment of the loading arrangement according to an advantageous embodiment of the invention,

(10) FIGS. 69-70—illustrate an example embodiment for adjusting the loading arrangement according to an advantageous embodiment of the invention,

(11) FIGS. 71-73—illustrate example embodiments of the loading arrangement according to an advantageous embodiment of the invention, and

(12) FIG. 74—illustrates an example method and embodiment for loading and/or unloading a load unit by the loading arrangement according to an advantageous embodiment of the invention.

DETAILED DESCRIPTION

(13) FIG. 1 illustrates a truck as an exemplary vehicle **103** with a loading arrangement **100** for loading and unloading load units **101** on and/or off a vehicle **103** according to an advantageous embodiment of the invention. The loading arrangement **100** comprises a frame **108**, which is mounted on the vehicle **103** in FIG. 1 advantageously in a detachable manner.

(14) Next number of embodiments of the invention are described in more details in connection to the FIGS. 2-74, where FIGS. 2-6 illustrate a principle of an exemplary transferring device **104** of the loading arrangement **100**,

(15) FIGS. 7-38 and also FIGS. 62-66 and FIG. 74 illustrate a principle of an operation of an exemplary loading arrangement **100** for loading and unloading load units **101** on and/or off the vehicle **103**, and FIGS. 39-49 illustrate a principle of an operation of an exemplary frame **108** for loading and unloading load units **101** on and/or off the vehicle **103** according to an advantageous embodiment of the invention.

(16) The frame **108** has first **108A** and second **108B** ends and is configured to be fastened to the vehicle in a detachable manner. The load unit **101** may be for example a container or sea container but is not limited to those only. The load unit **101** has first and second ends **101A**, **101B**, and advantageously comprises at least one fastening device. According to an example the load unit, such as the sea container, comprises at least two fastening devices in each corner of the load container. As can be seen in FIG. 17, for example, the sea container **101** comprises the fastening device in each corner of it, the fastening device having a hole in each side, the first fastening device **132** in the bottom portion and the second fastening device **102** in the side portion. The load unit **101** may naturally comprise also additional fastening devices and the inventions is not limited to these only.

(17) The frame **108** comprises advantageously two elongated frame elements **114**, **115** with first and second ends, having a longitudinal axis **117** between said ends **108A**, **108B**. In addition, the frame has advantageously an opening **118** between the elongated frame elements **114**, **115** so that for example the hook system **116** of the vehicle **103** can be operated between said frame elements **114**, **115** in the direction of said longitudinal axis **117**. For example, when the **108** frame is arranged on the vehicle **103**, the hook system **116** can be lifted up via the opening for loading or unloading the load unit **101** on or off the vehicle **103** so that there is no need to remove the frame, as can be seen e.g. in FIGS. 7-11.

(18) The frame **108** comprises advantageously also a dock **109**, which is arranged, such as fixed e.g. by welding or the like or fastened by a fastening means in a detachable manner, in the first end **108A** of the frame **108**. The dock **109** comprises advantageously a receiving portion **112** for receiving the transferring device **104** for at least loading and unloading the frame **108** on and/or off the vehicle by the transferring device **104** when the transferring device **104** is operated by the hook system **116** of the vehicle **103**. The transferring device **104** may be received and fastened to the dock **109** by a suitable securing device so that when the transferring device **104** is operated by the hook system **116** of the vehicle **103**, the frame **108** is also moved, such as loaded or unloaded to/from the vehicle (as can be seen e.g. in FIGS. 40-47).

(19) In addition, the dock **109** may also be configured to receive a first end **101A** of the load unit **101**, when the load unit **101** is loaded on the vehicle **103**. The receiving portion **112** may comprise advantageously also a locking device **113** to be fastened to a fastening device, such as to the first fastening device **132**, such as the first fastening hole **132** in the underside portion of the load unit **101** in the first end **101A**. In this way the first end **101A** of said load unit **101** can be fastened to the frame **108** in a detachable manner, and again via the frame **108** to the vehicle **103**, when the frame is fastened to the vehicle **103**. However, it is to be noted that the first end **101A** of said load unit **101** can be fastened to the vehicle also via the transferring device **104**, or that the load unit **101** can be just caught by the transferring device **104** (operated by the hook system **116** of the vehicle **103**) without specially securing the load unit to any fastening or securing device of the frame of vehicle.

(20) It is to be noted that the receiving portion **112** of the dock **109** may have two or more locking

devices **113** in different distances **136** from the first end **108A** towards the second end **108B** of the frame **108** for different sizes load units **101**, as can be seen e.g. in FIGS. **71, 72**. In this way also smaller or shorter load units **101** can be fastened via the locking devices **113** to the frame **108** and the vehicle **103** if they are wanted to be fastened to the frame and vehicle. According to an embodiment the distance **136** of the locking device from the first end **108A** is adjustable. The dock **109** may be provided e.g. suitable points for receiving the locking devices **113** in different distances. It is to be noted that the transferring device **104** can be kept fastened to the load unit **101** when the load unit **101** is loaded on the vehicle **103** and even if the load unit **101** is small or short load unit and it is not dragged to the first end **108A** of the frame **108**, such as is the case e.g. in FIG. **73**, or the transferring device **104** can be disconnected from the load unit **101**, whereupon the transferring device **104** can be dragged to the first end **108A** of the frame **108** and the load unit **101** can be secured to the frame **108** via a ramp **120** and its locking devices **124** or other securing or fastening solutions, for example, such as is the case e.g. FIGS. **71-72**.

(21) The frame **108** advantageously comprises also the ramp **120**, which is hinged **121** in the second end **108B** of the frame **108**. The ramp **120** is also configured to be hinged around a rear roller **122** of the vehicle **103**. This can be implemented e.g. by locking hinge pieces **133**, which are arranged around at least partly around the rear roller **122** and then fastened to the ramp **120**, advantageously to the second end portion of the ramp **120**, as can be seen e.g. in FIGS. **50-55**. The ramp **120** can be fastened to the vehicle **103** via said rear roller **122** in a detachable manner. In addition, it is to be noted that the locking hinge pieces **133** can be fastened to the ramp for example by bolts or the like, but also controllable locking hinge pieces **133** can also be used, such as pneumatically or hydraulically closable/openable locking hinge pieces **133**, which alleviates the assembling of the locking hinge pieces **133** and thus fastening and unfastening the ramp **120** and the frame **108** to the vehicle **103**. In addition, also an additional fastening device **142**, such as a rim screw, can be used for fastening the ramp and/or the hinge pieces **133**.

(22) It is to be noted that the ramp **120** is both hinged to the rear roller **122** of the vehicle **103** but also to the second end **108B** of the frame **108**, where the hinge point **121** in the second end **108B** of the frame **108** locates towards the first end **108A** from the rear roller **122**, when the frame **108** is mounted on the vehicle **103** and when the ramp is not tilted (such is the case e.g. in FIG. **10**). Thus, when the load unit **101** is loaded or unloaded and the ramp **120** is tilted around the both hinges **121, 122**, the frame moves slightly backwards (towards the second end **108B** so in direction of the longitudinal axis **117**). At the same the second end **108B** of the frame **108** moves slightly upwards in direction **134**, as can be seen in FIGS. **11** and **12**. It is to be noted also, that the frame **108** is advantageously fastened to the vehicle **103** only by hinging the ramp **120** around the rear roller **122** of the vehicle so that the movement of the frame during tilting the ramp **120** can be enabled. As can be seen e.g. in FIG. **56**, the frame **108** is secured to the vehicle **103** in a lateral direction **135** by introducing the elongated elements **114, 115** between securing fingers **119** of the vehicle **103**. The securing fingers **119** limit the movement of the frame **108** in the lateral direction **135** but allows the movement of the frame in the vertical direction **134** (FIG. **12**) being perpendicular to the lateral direction **135** and longitudinal axis **117**, and allows the movement of the frame **108** in the direction of the longitudinal axis **117**, when the ramp **120** is turned or tilted around the rear roller **122** and the hinge **121**.

(23) The ramp **120** may comprise guiding protrusions **123** for guiding the load unit in a lateral direction **135**, so preventing the load unit **101** moving in lateral direction **135**, as can be seen e.g. in FIGS. **25, 26** and **56**. In addition, the ramp **120** may comprise a locking device **124** to be introduced to an additional, e.g. a third, fastening device **125**, such as a hole **125** in the underside portion in the second end **101B** of the load unit **101**, as can be seen e.g. in FIGS. **28-30**. The locking device **124** of the ramp **120** is fastened to the third fastening device **125** in the second end **101B** of the load unit **101**, whereupon the load unit **101** is fastened to the frame **108** and again to the vehicle **103** also from the second end **101B** of the load unit.

(24) The arrangement **100** comprises in addition said transferring device **104** for catching and transferring the load unit **101** on and/or off the vehicle **103**. The transferring device **104** comprises advantageously a catching device **107** for catching the load unit **101** via the fastening device **102**, **132** of the load unit. The catching device **107** illustrated in Figures is a pin or plug, depending on the type of the fastening device of the load unit. For example, the fastening device of the typical sea container is a hole (at least two holes pointing different sides in each corner, so first and second fastening devices, as can be seen e.g. in FIG. 17), whereupon the catching device **107** is a plug. Naturally, the catching device **107** can be implemented also on other ways so that is it suitable for the fastening devices of the load unit in questions. In addition, according to an embodiment, the catching device **107** can be operated, such as closed and/or opened to/from the fastening device of the load unit remotely e.g. in hydraulically, pneumatically, electrically or the like manners known by the skilled person.

(25) The transferring device **104** comprises also a gripping point **131** (as can be seen in FIG. 57, 59-66, for example) to be gripped by the hook system **116** of the vehicle **103**. Thus, when the catching device **107** is fastened to the fastening device **102**, **132** of the load unit **101**, the load unit can be transferred, such as loaded or unloaded on or off the vehicle **103** by transferring device **104**. For transferring the load unit **101** the transferring device **104** is operated by the hook system **116** of the vehicle **103**. It is to be noted that in this way also the typical sea container can be loaded to the vehicle with the conventional type skip system, as can be seen e.g. in FIGS. 13-32.

(26) The dock **109** comprises a securing device **110** for securing the transferring device **104** to the frame **108** in a detachable manner. The exemplary securing device **110** in Figures (e.g. in FIGS. 2-4) is a hook or a rim screw, which can be fastened to a corresponding groove **111** in the transferring device **104**.

(27) FIGS. 2-6 and 59-66, for example, illustrate an exemplary transferring device **104**, wherein the transferring device **104** comprises two elongated arms **105**. At least one, preferably both, are joined to the transferring device **104** via a hinge **106** in a proximal end **105A** of the arms **105** so that the longitudinal axis **105C** of the two elongated arms form an angle (a). The longitudinal axis points advantageously downward angle. The angle between the arms is advantageously less than 180°, for example less than 130°, or less than 110° or more advantageously 90°, however depending on the implementation of the transferring device **104**.

(28) The arms **105** comprise the catching devices **107** advantageously in the distal ends **105B** of the arms. The catching devices **107** project advantageously inwards and towards the distal end of another elongated arm so towards the load unit **101** side and the fastening device **102**, **132** of the load unit in use, advantageously to the second fastening device **102** locating on the side of the load unit.

(29) The transferring device **104** has open and closed positions (see e.g. FIGS. 17 and 18). In the open position (FIG. 17) the angle (a) between the two elongated arms **105** is bigger than in said closed position (FIG. 18). In the open position the catching devices **107** are configured to be introduced to/from said fastening device **102**, **132** of the load unit **101** so that the transferring device **104** can be delivered around the sides of the first end **101A** of the load unit **101**. In the closed position the catching devices **107** are configured to be pushed/fastened or otherwise introduced into the fastening device **102**, **132** of the load unit, or locking holes **102**, **132** in the case of the sea container, as is the case in FIG. 18, for example.

(30) The exemplary the transferring device **104** comprises an operable connecting rod **126** arranged between the two elongated arms **105**, as can be seen in FIGS. 5, 6, 17-19, for example. The connecting rod **126** can be operated e.g. by the operating handle **138**. The connecting rod **126** is arranged to the distal portions **105B** of the arms so that the transferring device **104** can be opened to the opening position by separating the distal ends **105B** of the elongated arms **105** apart from each other by operating the connecting rod **126**. Correspondingly the transferring device **104** can be closed to the closed position by moving the distal ends **105B** of the elongated arms **105** towards

each other by operating the connecting rod **126**. The connecting rod **126** may be operated manually, pneumatically or hydraulically or even electric motors or servos can be used.

(31) The two elongated arms **105** of the transferring device **104** comprises advantageously also supporting projections **127**, as can be seen in FIG. **19**, for example. The supporting projections **127** are arranged in the distal ends **105B** of the arms **105** next to the catching devices **107**, projecting inwards and towards the distal end **105B** of another elongated arm **105** so towards the load unit **101** side in use. The supporting projections **127** are introduced to corresponding grooves **128** in the bottom portion of the load unit **101**. The supporting projections **127** support the load unit **101** from below portion of the load unit **101**, when the transferring device **104** is closed in the closed position, thus preventing for example lateral twist of the load unit **101** during loading the load unit **101** on the vehicle **103**.

(32) In addition, in some embodiments, such as illustrated e.g. in FIGS. **59-67** the transferring device **104** comprises in addition to said two elongated hinged arms **105** a gripping member **141** configured to grip the load unit in an opposite side, so in an upper side, than said two elongated hinged arms **105**, which are gripping the load unit in the lower or bottom side. The gripping member **141** may be fixed gripping member so that the opening angle is not adjustable, whereupon the gripping member **141** advantageously comprises catching devices which are configured to be introduced to upper side catching holes of the load unit (in the upper corners). In some embodiments the arms of the gripping member **141** may be implemented by telescopic arms so that the length of the arms of the gripping member **141** are adjustable for the load size. However, still the angle between the arms of the gripping member **141** may be fixed or adjustable.

(33) In addition, in some embodiments the transferring device **104** may also comprise a guiding boom **129**, as can be seen e.g. in FIGS. **10, 11, 13-16**. The guiding boom **129** is arranged or hinged to the upper portion of the transferring device **104** so that the free end of the guiding boom **129** is configured to be introduced to the upper portion of the load unit **101** when the transferring device **104** is moved towards the load unit **101**. The guiding boom **129** is configured to guide the transferring device **104** so that the longitudinal axis of said two elongated arms and a vertical axes of the transferring device **104** are essentially perpendicular to a longitudinal axis **101X** of the load unit **101**, as is the case in FIG. **14**, for example.

(34) The guiding boom **129** may also comprise a stopping member **130** in the free end of it, such as an outward projection so that when the guiding boom **129** guides the transferring device **104** in a vertical direction, the stopping member **130** prevents and stops the movements of the transferring device **104** in the vertical direction when the stopping member **130** meets the upper portion of the load unit **101**. In this way the guiding boom **129** targets the catching devices **107** at the right position of the fastening hole **102** of the load unit **101**.

(35) The transferring device **104** may also comprise or coupled with an elongated boom **139** having first and second ends **139A, 139B**, as can be seen in FIGS. **59-66**, for example. The elongated boom **139** is advantageously coupled with the transferring device **104** via a joint or levelling point **140**, which is arranged in the second end **139B** of the elongated boom **139**. The elongated boom **139** comprises also the gripping point **131** to be gripped by the hook system **116** of the vehicle **103**. The gripping point **131** is arranged next to the joint **140** to the second end **139B** of the elongated boom **139** and towards the first end **139A** of the elongated boom **139** from the gripping point **131**. The movement of the elongated boom **139** is limited around the joint **140** at least during loading and/or unloading the load unit **101** on or off the vehicle **103** by supporting the first end **139A** of the elongated boom **139** to the hook system **116** of the vehicle, advantageously to a boom of the hook system. Thus, when the joint or levelling point **140** locates closer to the second end **139B** of the elongated boom **139** than the gripping point **131**, it balances (due to gravity) a position and possible swinging of the transferring device **104** when it is moved e.g. towards the load unit **101** for catching it, such as is the case e.g. in FIG. **60**. This helps the operation of the transferring device **104**, because it does not swing so easily and is thus easier to couple with the load unit for catching

it. In addition, the elongated boom **139** functions as a lever arm and supports the first end **139A** (bottom portion of it) of the elongated boom **139** to the boom of the hook system **116** of the vehicle during loading/unloading the load unit **101** by the transferring device, as can be seen e.g. in FIGS. **60-66**. This allows more optimal position and angle to the hook system **116** for lifting the load unit during loading/unloading, namely the gripping point **131** can be arranged in a higher level in this way due to the elongated boom **139** and the mutual locations of the gripping point **131** and the joint **140**. Otherwise, the gripping point **131** will be in a lower level and beyond the dead angle of the hook system **116**, whereupon the lifting of the load unit e.g. during loading is easier and does not requires so much power of the hook system **116**.

(36) Thus, due to the relational arrangement of the gripping point **131** and the joint **140** the transferring device **104** can be balanced so that it does not swing so much when it is moved by the hook system **116** of the vehicle, which smoothens to settle the transferring device **104** to the load unit **101** for catching it.

(37) The arrangement comprises also a mechanical limiting member **143** for limiting the movement of the elongated boom **139** around the joint **140** during loading and/or unloading the frame **108** on or off the vehicle **103** by the transferring device **104**. Advantageously the limiting member **143** secures the first end **139A** of the elongated boom **139** to the frame structure **108** or to the dock **109** of the frame so that when the frame **108** is loaded/unloaded on or off the vehicle **103** by the transferring device **104**, the elongated boom **139** does not swing or move, as is the case in FIG. **59** where the frame **108** locates on the ground and the first end **139A** of the elongated boom **139** is secured to the frame structure **108** and after which the frame **108** can be e.g. loaded to the vehicle by operating the transferring device **104** by the hook system **116** of the vehicle. The limiting member **143** can be implemented e.g. by a suitable slot or groove in the first end **139A** of the elongated boom **139** so that the slot or groove meets the corresponding shoulder or the like in the dock **109** or other portion of the frame **108**.

(38) Still in addition, the arrangement may comprise an adjusting member **144** for reduce or expand the inner size of the elongated boom **139** so that it can be fitted tightly around the different size boom of the hook systems **116**, as can be seen in FIG. **61**, for example. According to an example the adjusting member **144** can be provided by a spring element or with flexible rubber element so to fit the adjusting member **144** better and tightly against the boom of the hook systems **116** and also thereby relieving mechanical stresses and torques caused by the relational movement of the elongated boom **139** and the hook systems **116**, for example.

(39) The operation of the arrangement **100**, in particularly the frame **108** and exemplary transferring device **104** can be seen in Figures, and in particularly in FIGS. **7-32**, **59-66** and **73**, wherein e.g. in FIG. **7** the transferring device **104** is gripped by the hook system **116** of the vehicle **103**. The hook system **116** is gripped around the gripping point **131**, whereupon the transferring device **104** can be moved backwards (when the securing device **110** of the dock **109** is opened, as is the case e.g. in FIG. **3**) and towards the load unit **101** lying on the ground, as can be seen e.g. in FIGS. **8-11** and **62-63**. The movement of the hook system **116** is done in the opening **118** between the elongated frame elements **114**, **115**.

(40) When the transferring device **104** reaches the first end **101A** of the load unit, the guiding boom **129** (if used) meets the upper portion of first end **101A** and when the transferring device **104** is moved further, so downwards, the guiding boom **129** guides the transferring device **104** in the right position (as can be seen in FIGS. **10**, **11**, **13**, **14**) and the stopping member **130** finally prevents and stops the movements of the transferring device **104** (as can be seen in FIGS. **15**, **16**) so that the catching devices **107** of the transferring device **104** is at the right position in relation to the fastening hole **102** of the load unit **101**. It is to be noted that when the transferring device **104** is moved to the first end **101A** of the load unit **101**, the transferring device **104** must be in open position so that it can be moved around the sides of the load unit **101**, as is the case in FIG. **17**. When the transferring device **104** and the catching devices **107** are in the right position, the

transferring device **104** can be closed to the closed position, as is the case in FIG. **18**. As can be seen in FIGS. **19, 20**, the supporting projections **127** are also introduced to corresponding grooves **128**, when the transferring device **104** is closed to the closed position.

(41) When the transferring device **104** is moved backwards, it is configured to meet also the ramp **120** and tilt to the position for receiving the load unit **101** to be loaded on the vehicle **103**, as is the case in FIG. **12**.

(42) Now, when the transferring device **104** is closed to the closed position and caught the first end **101A** of the load unit **101** (FIGS. **18** and **63**, for example), the load unit **101** can be transferred by the transferring device **104** by operating the hook system **116** of the vehicle **103**, as can be seen in FIGS. **21-32** and **63-66**. When the hook system **116** is retracted forwards, the transferring device **104** pulls the load unit **101** so that the load unit **101**, advantageously the bottom portion of the load unit **101**, meets the tilted ramp **120**, as is the case in FIGS. **23** and **64-65**. When the load unit **101** is dragged further, it travels along the surface of the ramp **120**, and at the same the ramp **120** delivers the load or stress to the rear rollers **122** of the vehicle **103**. The guiding protrusions **123** of the ramp **120** guides the load unit **101** and prevents the load unit **101** moving in lateral direction **135**, as can be seen in FIGS. **25-27**, for example.

(43) When the hook system **116** and thus the transferring device **104** are moved further (e.g. FIGS. **27-29** and **65-66**), the transferring device **104** and the first end **101A** of the load unit **101** reaches finally the receiving portion **112** of the dock **109**, transferring device **104** can then be fastened to the dock via the securing device **110** and also the first end **101A** of the load unit **101** can then be fastened by the locking device **113** (e.g. FIG. **29**), if used. It is to be noted that technically the fastening or securing of the transferring device **104** to the dock and/or the load unit **101** by the locking device **113** is not mandatory, but of course it increases safety. In addition, the second end **101B** of the load unit **101** can be fastened by the locking device **124** of the ramp **120**, which can be fastened to the fastening device **125**, such as the third hole **125** of the sea container, in the underside portion in the second end **101B** of the load unit **101**, as is the case in FIG. **30**. The unloading the load unit **101** off the vehicle **103** can be done in reversed way than loading.

(44) It is to be noted that the load unit **101** can also be caught by the transferring device **104** also from on another trailer or transferred to another trailer, as can be seen in FIG. **74** and not only from the ground.

(45) It is to be noted also that the vehicle **103** can be moved normally by driving with the load unit **101** on it. In addition, the load unit **101** can be removed off the vehicle **103** for example by a separate crane **137**, as is the case in FIGS. **33** and **34**. This can be enabled by opening the locking device **113** of the dock **109**, locking device **124** of the ramp **120** as well as the catching devices **107** of the transferring device **104** and opening the transferring device **104** to the opening position.

(46) In addition, it is to be noted, that the load unit **101** can be unloaded off the vehicle also with the frame **108**, as is illustrated in FIGS. **35-38** and **67**. For this, the locking hinge pieces **133**, which are arranged around at least partly around the rear roller **122**, must be opened, such as unfastened from the ramp **120** and removed around the rear roller **122**, whereupon the frame **108** with the load unit **101** can be moved. The unloading can be done by operating with the hook system **116** of the vehicle **103**, so moving the hook system **116** and thus the transferring device **104** backwards. In this the transferring device **104** is caught by the hook system **116** via the gripping point **131**. It is to be noted that when unloading the frame **108** with the load unit **101**, the locking device **113** of the dock **109** and locking device **124** of the ramp **120** are advantageously locked to the corresponding points in the load unit **101**, as is discussed elsewhere in this document. In addition, also the catching device **107** of the transferring device **104** is also fastened the fastening device **102** of the load unit **101**.

(47) When the frame **108** together with the load unit **101** is unloaded and landed to the ground, the catching device **107** of the transferring device **104** can be opened from the fastening device **102**, **132** of the load unit **101**, whereupon the transferring device **104** can be lifted away by the hook

system **116** of the vehicle **103**, as can be seen in FIG. **38**. Then the frame **108** can be left to the ground with the load unit **101**. The transferring device **104** can be lifted away by the hook system **116** of the vehicle **103**, when the catching device **107** of the transferring device **104** is opened. It is to be noted that the frame **108** can be transferred with the load unit also on another trailer or vehicle, as is shown in FIG. **74**, whereupon also that trailer or vehicle can be prepared or enabled to carry the load unit, such as a sea container, even if it was not suitable for carrying the load unit, such as a sea container before the frame **108**. This is also clear advantage of the present invention.

(48) FIGS. **39-57** illustrate an exemplary principle of loading and mounting the frame on the vehicle **103**. In FIG. **39** the frame **108** locates on the ground and the transferring device **104** is moved towards the frame **108** by the hooks system **116**. The transferring device **104** is advantageously fastened or secured to the dock **109** by the securing device **110** of the dock **109**, after which the frame **108** can be loaded on the vehicle **103** by gripping the transferring device **104** secured to the dock **109** and frame **108** by the hook system **116** and dragging the transferring device **104** together with the frame **108** on the vehicle **103**, as can be seen in FIGS. **41-49**. When the frame **108** is loaded on the vehicle **103**, the second end **108B** of the frame **108** can then be fastened to the rear portion of the vehicle **103** via the rear rollers **122** of the vehicle **103** by the locking hinge pieces **133** of the ramp **120**, as is described elsewhere in this document and as can be seen e.g. FIGS. **50-55**. The securing fingers **119** limits the movement of the frame **108** in the lateral direction **135**, and optionally the transferring device **104** can be kept fastened to the dock **109** via the securing device **110** of the dock **109**, as well as the hook system **116** may be kept gripped to the gripping point **131** of the transferring device **104**, whereupon the frame **108** is fastened to the vehicle **103** both via the rear roller **122** as well as via the transferring device **104** and the hooks systems **116** of the vehicle **103**. Also, an additional fastening device **142**, such as a rim screw, can be used for fastening the ramp and/or the hinge pieces **133**. However, it is to be noted that the securing device **110** of the dock **109** can be opened, as well as release the hook system **116** from the gripping point **131**, and still the frame **108** (also with the load unit **101**) is secured to the vehicle **103** via the rear rollers **122**. The frame **108** can be removed off the vehicle **103** only opening and removing also the locking hinge pieces **133** of the ramp **120**.

(49) FIGS. **69-70** illustrate an example embodiment for adjusting the loading arrangement according to an advantageous embodiment of the invention for different size hook systems **116**. It is to be noted that the size, such as length **146** of the distal boom of the hook system **116** may vary, whereupon the frame **108** comprises an adjusting member **145** for adjusting the position, or height **147**, of the first end **108A** of the frame **108** (and thus the position of the dock and receiving portion **112** for receiving the transferring device **104**) from the body of the vehicle **103** and thus fitting the position of the gripping point **131** of the transferring device **104** in relation to the hook system (especially the distal end of it) **116** when the transferring device **104** is transferred (with or without the load unit) on to the vehicle **103**.

(50) The invention has been explained above with reference to the aforementioned embodiments, and several advantages of the invention have been demonstrated. It is clear that the invention is not only restricted to these embodiments but comprises all possible embodiments within the spirit and scope of the inventive thought and the following patent claims.

(51) For example, it is to be noted that in addition to the disclosure and examples, the load units may be also other kinds of load units than container or sea container. In addition, it is to be noted that the transferring device **104** illustrated in FIGS. **1-53**, and **55-56** is only example and also other kind of transferring devices can be used, such as an exemplary transferring device **104X** illustrated in FIG. **54**, where the transferring device **104X** still comprises the gripping point **131** to be gripped by the hooks system **116** of the vehicle **103**. In addition, the transferring device **104X** comprises also the catching device **107** to be fastened to the fastening device **102**, **132** of the load unit **101**. However, it is to be noted that the catching device **107** may be for example a hook pin, pivot, plug or cable to be introduced to a corresponding fastening device of the load unit. In addition, it is to be

noted that the corresponding fastening device of the load unit may be e.g. a hole, groove, hook or even a gripping point like the gripping point **131** of the transferring device **104**. In addition, as is described, the dock may comprise said securing device for securing the transferring device in a detachable **10** manner but said securing device can be provided also elsewhere in said first end of the frame, for example.

(52) The features recited in dependent claims are mutually freely combinable unless otherwise explicitly stated.

Claims

1. A loading arrangement for loading and unloading load units on and off a vehicle, said load unit having first and second ends, wherein the arrangement comprises a transferring device for catching and transferring the load unit on and off the vehicle and a frame having first and second ends and to be fastened to the vehicle in a detachable manner, wherein: the frame comprises a dock, which is arranged in the first end of the frame and wherein the dock comprises a receiving portion for receiving the transferring device for at least loading and unloading said frame on and off the vehicle by the transferring device when said transferring device is operated by a hook system of the vehicle, and said transferring device comprising a gripping point to be gripped by the hook system of the vehicle and a catching device for catching said load unit and for transferring it on or off the vehicle, when the transferring device is operated by said hook system of the vehicle, wherein said transferring device comprises: two elongated arms, at least one arm joined to said transferring device via a hinge, where said hinge is arranged in a proximal end of the at least one arm, whereupon longitudinal axes of said two elongated arms form an angle, said arms having said catching device arranged in distal ends of the arms, and wherein said transferring device has open and closed positions, wherein in said open position the angle between the two elongated arms is bigger than in said closed position and wherein in said open position said catching device is configured to be introduced to/from the load unit and wherein in said closed position said catching device is configured to be fastened to the load unit, thereby allowing catching and transferring the load unit.

2. The loading arrangement of claim 1, wherein the frame comprises: a ramp, which is hinged in the second end of the frame, and wherein the ramp is configured to be turned around the hinge for receiving and guiding the load unit on and off the vehicle, when the load unit is transferred on or off the vehicle, and wherein said ramp is in addition configured to be hinged around a rear roller of the vehicle and thereby divide load of the load unit to the vehicle structure via the rear roller of the vehicle, when said load unit is transferred, and in addition thereby to fasten said frame to the vehicle via said rear roller in a detachable manner.

3. The loading arrangement of claim 2, wherein the ramp comprises guiding protrusions for guiding said load unit in a lateral direction and a locking device to be introduced to an additional fastening device of the load unit in said second end and to be fastened to said additional fastening device, thereby fastening said second end of said load unit to said frame, when said load unit is loaded on the vehicle.

4. The loading arrangement of claim 1, wherein the frame comprises two elongated frame elements with first and second ends, said frame having a longitudinal axis between said first and second ends and having an opening between the elongated frame elements so that the hook system of the vehicle can be operated between said frame elements in said opening in the direction of said longitudinal axis, for loading or unloading the load unit on or off the vehicle.

5. The loading arrangement of claim 1, wherein said frame is configured to receive and secure said transferring device in a detachable manner, and wherein the transferring device is configured to be released for transferring the load unit on or off the vehicle and to be secured for loading and/or unloading said frame on or off the vehicle when said transferring device is operated by a hook

system of the vehicle or to be secured for transporting said transferring device.

6. The loading arrangement of claim 1, wherein the dock comprises a securing device for securing said transferring device in a detachable manner, and wherein said securing device comprises a hook, pin, pivot, plug or cable to be introduced to a corresponding groove in said transferring device.

7. The loading arrangement of claim 1, wherein the catching device comprises a hook, pin, pivot, plug or cable, and wherein the catching device is configured to be fastened to a second fastening device of the load unit so that it allows fastening and unfastening a locking device of the dock to a fastening device of the load unit.

8. The loading arrangement of claim 1, wherein the catching device is arranged to project towards the distal end of another elongated arm, and wherein said catching device is configured to be fastened to fastening device of the load unit, and wherein in said open position said catching device is configured to be introduced to/from the fastening device of the load unit and wherein in said closed position said catching device is configured to be fastened to the fastening device of the load unit.

9. The loading arrangement of claim 1, wherein the transferring device comprises an operable connecting rod arranged between the two elongated arms and attached to the distal portions of the arms, said connecting rod being configured to open said transferring device to the opening position by separating the distal ends of the elongated arms apart from each other and to close said transferring device to the closed position by moving the distal ends of the elongated arms towards each other.

10. The loading arrangement of claim 1, wherein the two elongated arms of the transferring device comprises also supporting projections arranged in said distal ends of the arms and projecting towards the distal end of another elongated arm, said supporting projections being configured to be introduced to corresponding grooves in the bottom portion of the load unit and thereby supporting the load unit from below portion of the load unit, when said transferring device is in said closed position.

11. The loading arrangement of claim 1, wherein the arrangement comprises an elongated boom having first and second ends, where the elongated boom is coupled with the transferring device via a joint arranged in the second end of the elongated boom and wherein the elongated boom comprises also said gripping point to be gripped by the hook system of the vehicle, said gripping point being arranged next to said joint to the second end of the elongated boom and towards the first end of the elongated boom from the gripping point.

12. The loading arrangement of claim 11, wherein the movement of said elongated boom is limited around the joint during loading or unloading the load unit on or off the vehicle by supporting the first end of the elongated boom to the hook system of the vehicle.

13. The loading arrangement of claim 11, wherein the arrangement comprises a mechanical limiting member for limiting the movement of the elongated boom around the joint during loading or unloading the frame on or off the vehicle by the transferring device.

14. The loading arrangement of claim 1, wherein said transferring device comprises in addition to said two elongated hinged arms a fixed gripping member configured to grip the load unit in an opposite side than said two elongated hinged arms.

15. The loading arrangement of claim 1, wherein said transferring device comprises a guiding boom, which is arranged to the upper portion of the transferring device, and wherein a free end of said guiding boom is configured to be introduced to the first end of the load unit, when the transferring device is moved towards the load unit to be caught and transferred, wherein said guiding boom is configured to guide the transferring device so that the longitudinal axis of said two elongated arms and a vertical axes of the transferring device are essentially perpendicular to a longitudinal axis of the load unit to be caught.

16. The loading arrangement of claim 15, wherein said guiding boom comprises a stopping

member in the free end of it, whereupon the guiding boom is also configured to guide the transferring device in a vertical direction so that when the stopping member of the free end of the guiding boom meets the upper portion of the load unit, the stopping member configured to stop the movement of the transferring device in said vertical direction and thereby targets the catching devices at the position of a fastening hole of the load unit.

17. The loading arrangement of claim 1, wherein a locking device of the dock is configured to be fastened to the fastening device of the load unit in said first end and fastening said first end of said load unit to said frame in a manner for enabling said transferring device to be opened to said open position or in an unsecured position so to remove said transferring device away from the dock and frame.

18. The loading arrangement of claim 1, wherein the frame is configured to be secured to the vehicle in a lateral direction, perpendicular to said longitudinal axis, via elongated elements so that said elongated elements are configured to be introduced between securing fingers of the vehicle and thereby secure said frame in the lateral direction but allowing said frame to be moved in a vertical direction being perpendicular to said lateral direction and longitudinal axis, and allowing said frame to be moved in the direction of said longitudinal axis, when the ramp is turned around a rear roller and the hinge.

19. A method for loading and unloading load units on and off a vehicle, said load unit having first and second ends, the method comprising: providing a frame that has first and second ends and a dock in the first end for receiving a transferring device, whereupon the frame is loaded and unloaded on and/or off the vehicle by the transferring device when said transferring device is secured to said dock and operated by a hook system of the vehicle, and wherein said frame is fastened to the vehicle in a detachable manner; providing the transferring device used for catching and transferring the load unit on or off the vehicle, the transferring device comprising: a catching device, which is used for catching said load unit and for transferring it on or off the vehicle when the transferring device is operated by said hook system of the vehicle; and two elongated arms, at least one arm joined to said transferring device via a hinge, where said hinge is arranged in a proximal end of the at least one arm, whereupon longitudinal axes of said two elongated arms form an angle, said arms having said catching device arranged in distal ends of the arms; and a gripping point, which is gripped by the hook system of the vehicle, wherein said transferring device has open and closed positions, wherein in said open position the angle between the two elongated arms is bigger than in said closed position and wherein in said open position said catching device is configured to be introduced to/from the load unit and wherein in said closed position said catching device is configured to be fastened to the load unit, thereby allowing catching and transferring the load unit; and catching the load unit and transferring the load unit on and off the vehicle by the transferring device.
